

Nutella Intake

Data-driven approach in daily jar-to-mouth delivery flow. An end-to-end pipeline.

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Background

My neighbour loves Nutella. And I mean LOVES Nutella - every single morning, is a Nutella brekkie morning. With a study subject like that, it was impossible not to look at the habit through the lens of data.

I asked my subject to weigh the jar every day after breakfast. The measurements were first recorded by hand on paper, then transferred into Excel, with each jar stored on a separate sheet to keep the data organised and future-proof.

With five weeks of daily observations in hand, the analysis could begin.

Data Collection

The workflow was deliberately simple:

1. Weigh the jar after breakfast.
2. Write down the reading on paper.
3. Transfer the values into Excel.
4. Clean, analyse, and visualise the results in Python.
5. Design a poster.

Two jars were used during the tracking period. The starting weight of each new jar was recorded, but these rows were excluded from the consumption analysis because they do not represent actual intake.

Data Preparation

The raw workbook contains two sheets, one per jar. These were merged into a single dataset, cleaned, and annotated with weekday and week number.

```
sheet1 = pd.read_excel("../Nutella.xlsx", sheet_name=0)
sheet2 = pd.read_excel("../Nutella.xlsx", sheet_name=1)

df = (pd.concat([sheet1, sheet2], ignore_index=True)
      [["Day", "weight", "daily consumption"]]
      .iloc[:40]
      .drop([0, 27])
      .rename(columns={"Day": "c_day", "daily consumption": "daily_consumption"})
      .astype({"weight": int, "daily_consumption": int})
      .reset_index(drop=True))
```

The final cleaned dataset contains 35 daily observations across 5 full weeks.

First Look at the Data

	c_day	weight	daily_consumption	dow	week	row_index
0	1	1427	34	Monday	1	0
1	2	1384	43	Tuesday	1	1
2	3	1343	41	Wednesday	1	2
3	4	1314	29	Thursday	1	3
4	5	1281	33	Friday	1	4
5	6	1246	35	Saturday	1	5
6	7	1194	52	Sunday	1	6

```
df[['daily_consumption', 'weight']].describe().round(1)
```

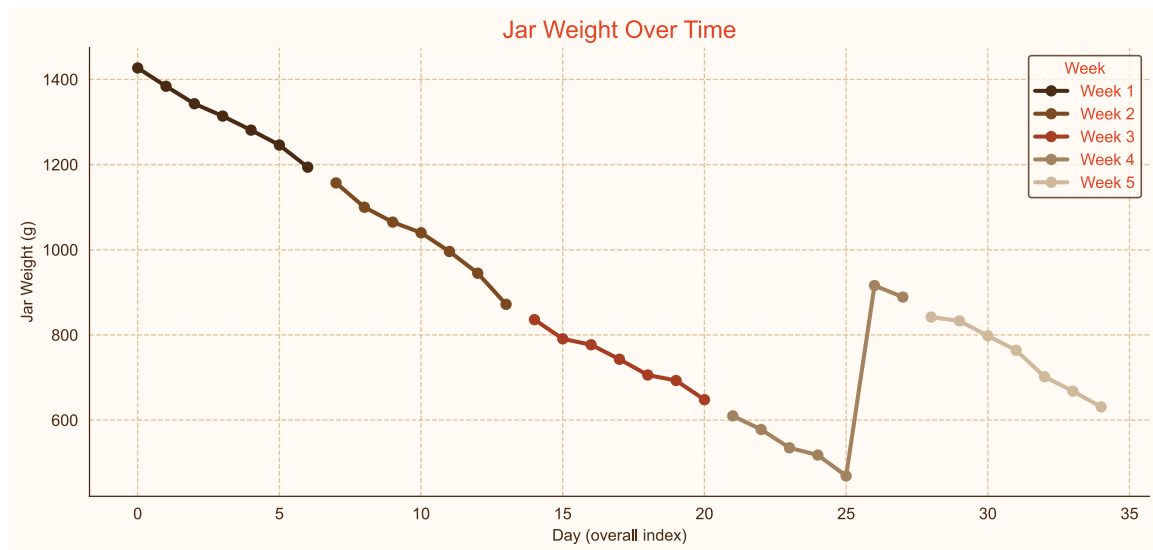
	daily_consumption	weight
count	38.0	38.0
mean	39.7	864.6
std	14.3	277.9
min	9.0	455.0
25%	34.0	653.0
50%	37.0	815.5

	daily_consumption	weight
75%	48.5	1058.8
max	73.0	1427.0

The average daily consumption was about 38g, but the spread was wide enough to suggest that some days were clearly heavier than others.

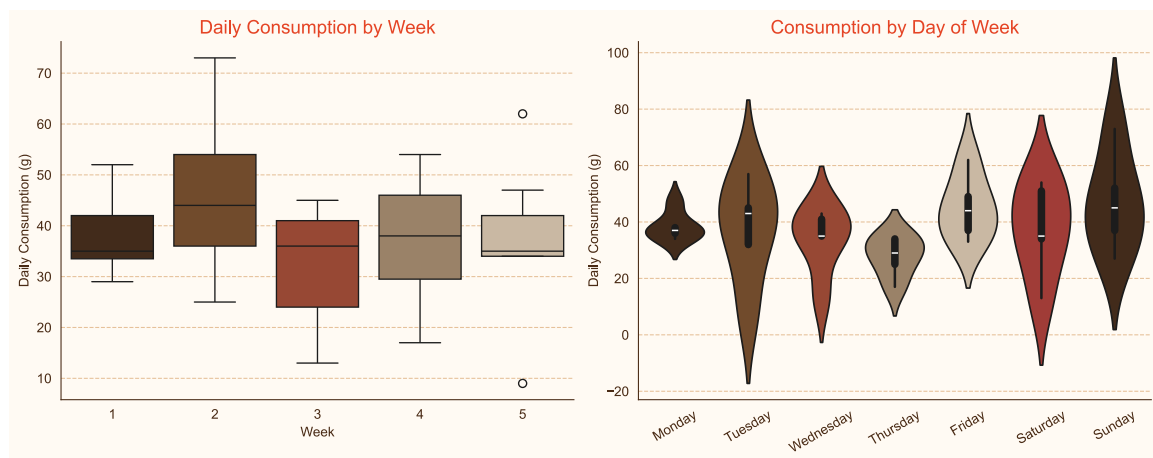
Jar Weight Over Time

Each coloured line below tracks how the jar weight dropped over time. This gives the most intuitive picture of consumption: steeper declines mean heavier breakfast activity.



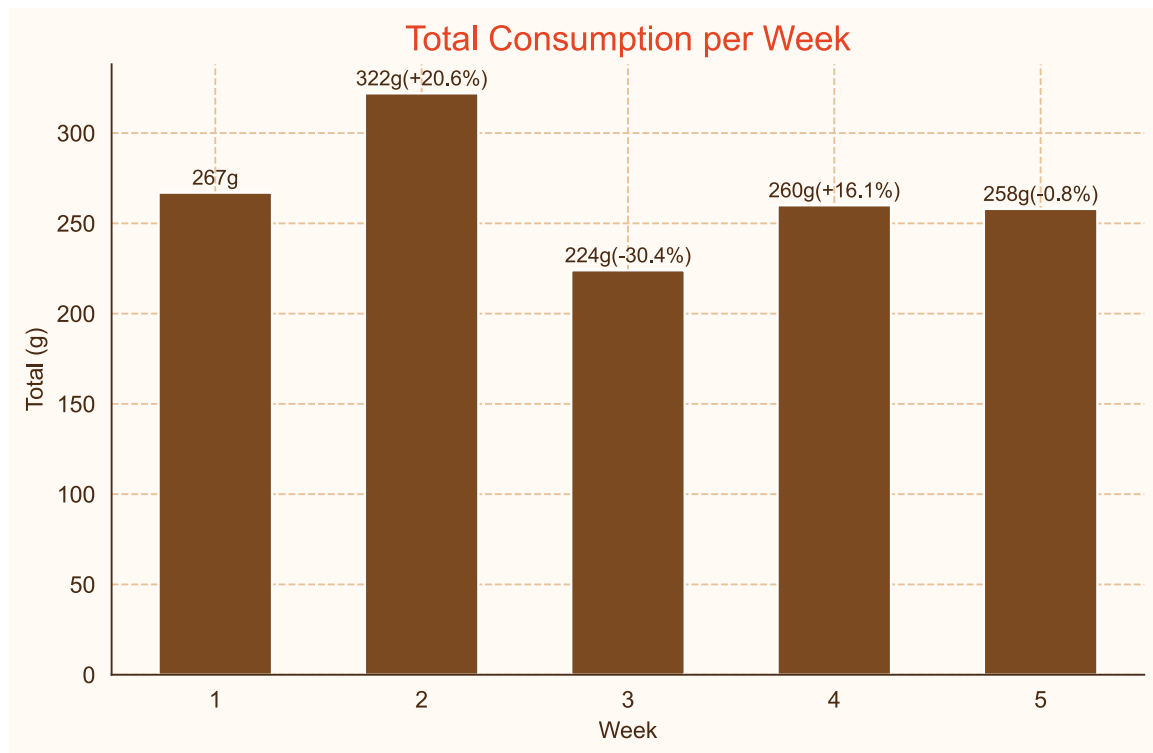
Variation in Consumption

The distribution plots below show that consumption varied not only from week to week, but also by day of the week.



Weekly Performance

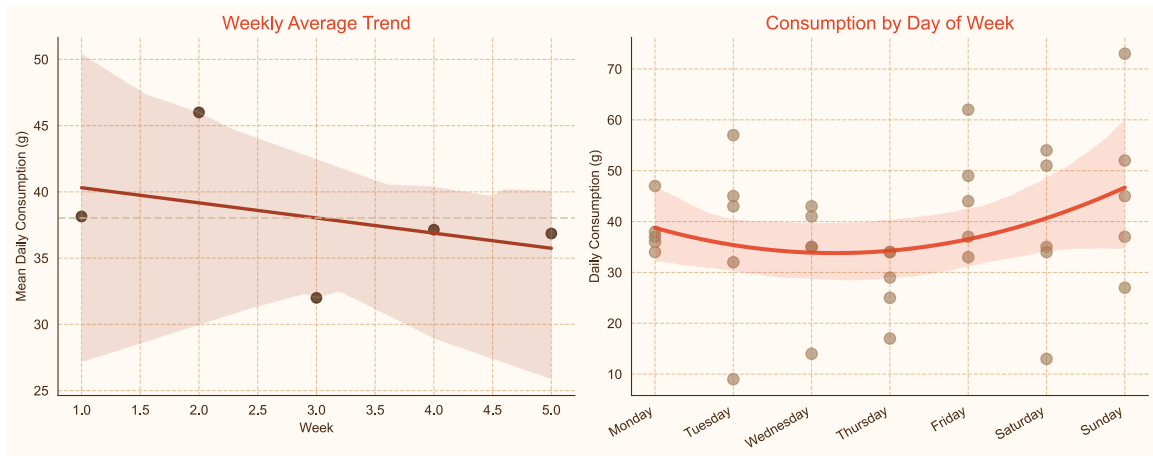
	Week	Total (g)	Avg/Day (g)	Δ vs Prev Week (%)
0	1	267	38.1	NaN
1	2	322	46.0	20.6
2	3	224	32.0	-30.4
3	4	260	37.1	16.1
4	5	258	36.9	-0.8



Week 2 was the peak period, while Week 3 showed the sharpest drop. Even in a playful project, this kind of weekly contrast is exactly the sort of pattern that makes longitudinal data interesting.

Trends Across Time

To test whether the pattern was changing over time, two regressions were used: a linear regression over weekly averages, and a quadratic fit over day-of-week values.

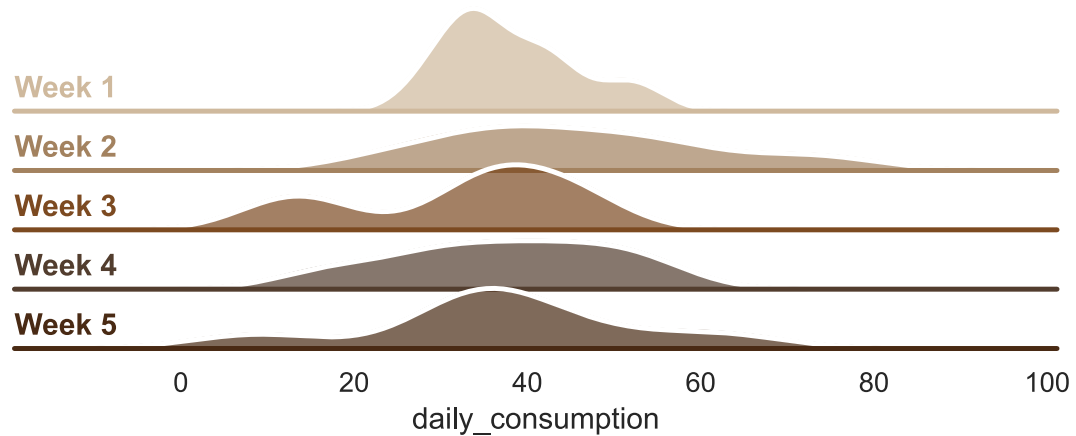


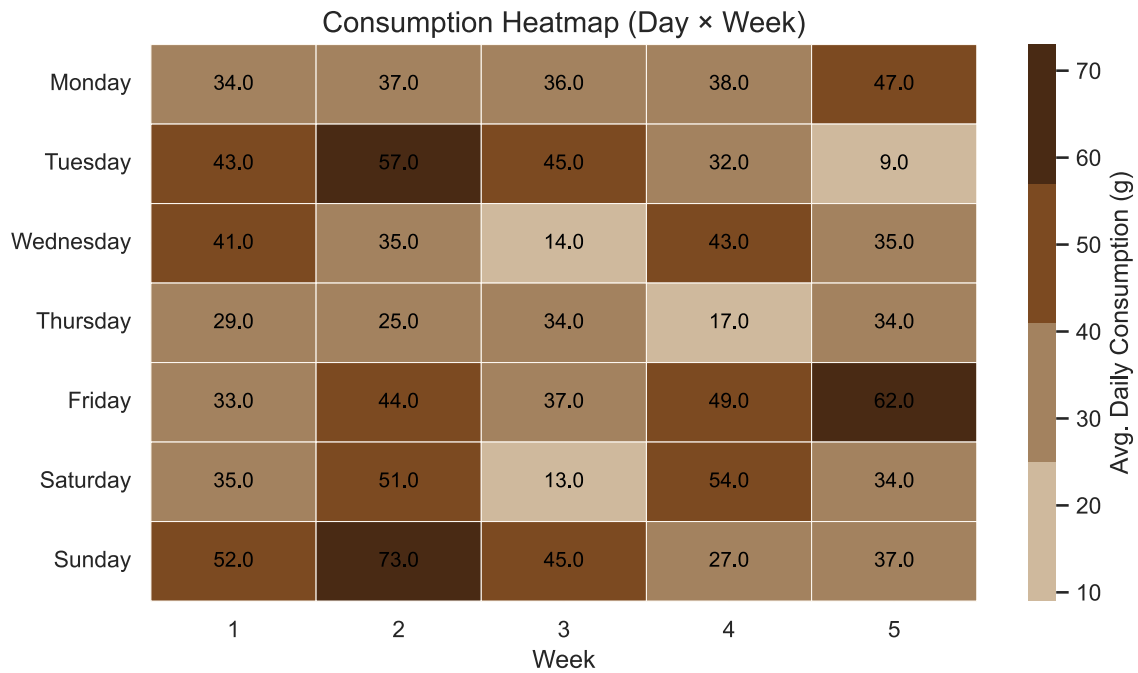
The clearest rhythm appears across the week itself: lower values around the middle of the week, and higher values heading into the weekend. The high consumption in Week 2 is followed by a lower than average consumption in Week 3. If these two are seen as related and largely offsetting, the average of weekly mean remains remarkably stable over the test period - suggesting a consistent baseline over time.

Density and Heatmap Views

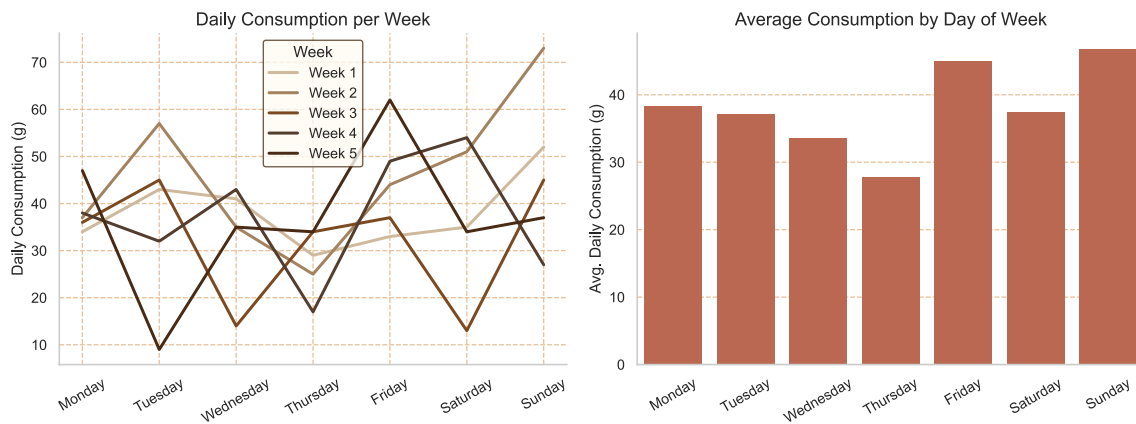
Different chart types reveal different aspects of the same behaviour. The ridge plot makes the distribution shape visible, while the heatmap highlights where the strongest day-week combinations occurred.

Nutella Consumption Density by Week





Week Profiles and Day Averages



The average-by-day view makes one result especially clear: Sundays were the strongest Nutella days.

Key Findings

Metric	Value
Total consumption	1,331g over 35 days
Average daily consumption	~38g
Peak week	Week 2 — 322g
Lowest week	Week 3 — 224g

Metric	Value
Highest single day	Sunday, Week 2 — 73g
Peak day of week	Sunday
Max vs Min daily ratio	8.1×

- The Sunday Spike: The heatmap and day-of-week plots reveal a consistent pattern: Sunday is the undisputed “heavy-intake” day. This is likely tied to the leisure time of a weekend, where breakfast isn’t rushed and serves as a relaxed, multi-session ritual.
- Mid-week Dip: Consumption is noticeably lower from Tuesday through Thursday. This creates a “u-shaped” weekly profile, where the routine of the working week seems to exert a disciplining effect on Nutella intake compared to the high-freedom weekends.
- Week 2 Anomaly: Week 2’s consumption spike (322g) is the clear outlier. The ridge plot for that week is notably flatter and shifted to the right, confirming that this wasn’t just one high day, but a generally more indulgent week.
- The Variability Ratio: An 8.1× ratio between minimum and maximum daily consumption is quite high for a daily habit. It highlights that while Nutella is a daily staple, the quantity consumed is highly elastic, reacting strongly to the day of the week or perhaps the specific mood of the morning or other external factors.

Fun Facts

A light project still allows for unnecessary but enjoyable extrapolation.

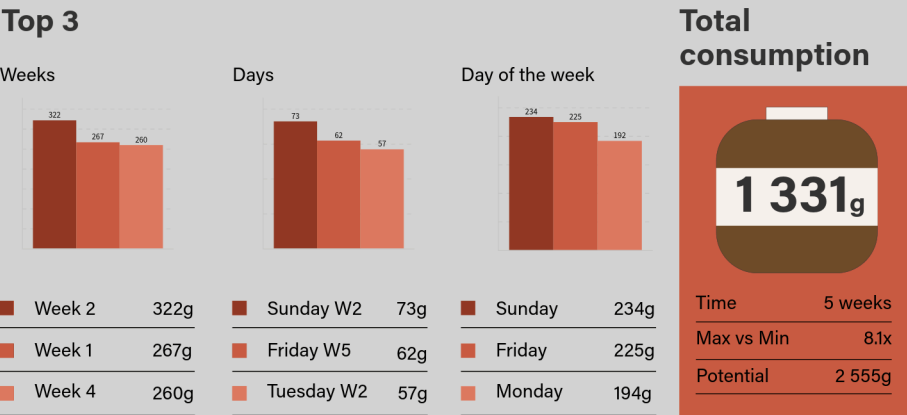
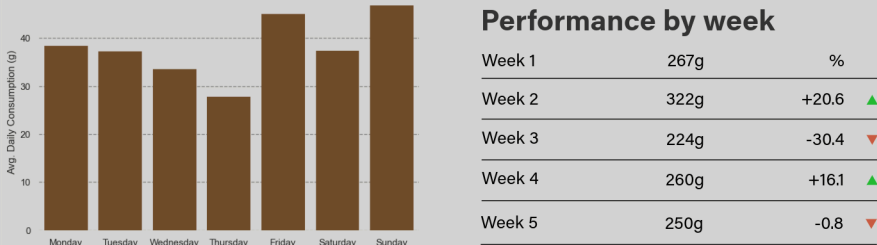
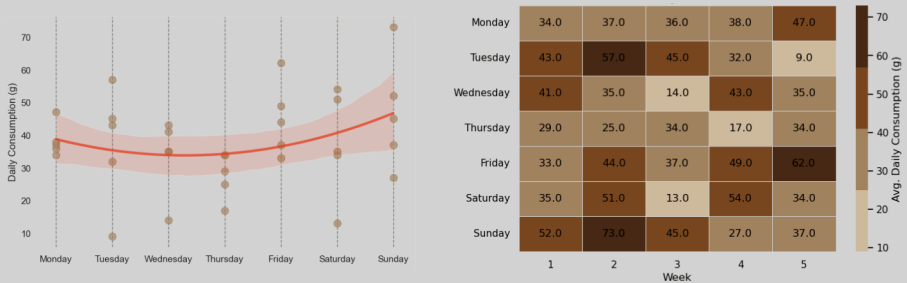
```
avg = avg_daily
elephant_g = 6_000_000
days_elephant = elephant_g / avg
years_elephant = days_elephant / 365.25
```

Fun Fact	Answer
Years to eat an elephant’s weight in Nutella	~432,000 years
Grams per minute on a Sunday (spread over 24h)	0.033 g/min
1kg jars needed for 100 years	~1,389 jars

Final Poster

The project also produced a single-page poster designed to summarise the full dataset at a glance.

Nutella intake



A spoon of hazelnut spread a day, keeps the doctor away

Figure 1: Nutella Intake Poster

Reflections

This project takes the formal data pipeline and spreads it to an informal scenario. A real-world mini project like this is always fun.

This project might have started as a joke, but with a serious report like this - it graduated to be a fun little mini project - an end-to-end joke? backed by data?

This project might have started as a joke, but with a serious report like this - is it still one? Yes... but a joke backed by data! - whatever that means. Oh, and it doesn't stop there - it also is an end-to-end joke. :)

A few things stood out:

- **Data collection design matters.** The more control one has over the data collection, the easier it is to then work with the data.
- **Cleaning decisions matter.** The first weight of each new jar had to be removed so the analysis reflected actual intake rather than stock resets. This follows the previous point - the more you know about the data, the easier it is to make decisions about the data.
- **Presentation matters.** A notebook is useful for analysis, but to - get the data out of the kitchen and into the living room - a report and poster make it more understandable to everyone else.

The most enjoyable part of the project is that it takes something completely ordinary - breakfast habits - and treats it as worthy of analysis. That is a good reminder that tasty data is often hiding in plain cupboard - waiting to be opened and spread across a thin pdf report.